MA1521 Calculus for Computing

Lecture Group 2

Lecturer: Prof To Wing Keung

Lesson 19: Tuesday 20 October 2026

(Week 10)

Remark. Q019.1 to Q19.6 are on Section 14.5 of Thomas' Calculus.

Q19.7 is on Section 14.8 on change of variable formula for triple integrals.

Q19.1 Evaluate the triple iterated integral

$$\int_0^1 \int_0^2 \int_0^3 (xy + 2z)\, dz\, dy\, dx.$$

Q19.2 Consider the region in the xyz-space given by

$$D = \{(x, y, z) : 0 \leq x \leq 3,\ \ 1 \leq y \leq 3,\ \ 0 \leq z \leq 4\}.$$

Evaluate the triple integral $\iiint_D (x^2 + xyz)\, dV$ by computing an iterated integral in the order of $dz\, dy\, dx$.

Q19.3. Evaluate the triple integral in **Q19.2** again by computing an iterated integral in the order of $dx\, dy\, dz$.

Q19.4. Consider the region in the ***xyz***-space given by

$$D = \{(x, y, z) \ : \ 0 \leq x \leq 1, \ \ 0 \leq y \leq 1 - x^2, \\ 3 \leq z \leq 4 - x^2 - y\}.$$

Evaluate $\iiint_D x \, dV$.

Q19.5. Consider the solid region in the ***xyz***-space given by

$$D = \{(x, y, z) \ : 0 \leq y \leq 1, \ \ 0 \leq x \leq \sqrt{4 - y^2}, \\ 0 \leq z \leq 2x + y\}.$$

Find the volume of D.

Q19.6. Consider the region W in the first octant of the xyz-space bounded by the coordinate planes $x = 0$, $y = 0$, $z = 0$, the plane $y = 1$ and the plane $x + z = 1$. Let $g(x, y, z)$ be a function defined on W.

(i) Write the triple integral $\iiint_W g(x, y, z)\, dV$ as an iterated integral with integration order $dz\, dx\, dy$.

(ii) Write the triple integral $\iiint_W g(x, y, z)\, dV$ as an iterated integral with integration order $dy\, dx\, dz$.

Q19.7. Consider the region in the ***xyz***-space given by

$$D = \{(x, y, z) \,:\, 1 \leq x \leq 2,\ 0 \leq xy \leq 2,\ 0 \leq z \leq 1\}.$$

Evaluate

$$\iiint_D (x^2 y + 3xyz)\, dx\, dy\, dz$$

by applying the transformation

$$u = x, \quad v = xy, \quad w = 3z$$

and integrating over an appropriate region in the ***uvw***-space.